Fiscal Year:	FY 2021	**Task Last Updated:**	FY 01/28/2022
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Project Title:	Automated Vascular Access for Spaceflight		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	(1) **ExMC**:Exploration Medical Capabilities		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2019 HERO 80JSC019N0001-FLAGSHIP & OMNIBUS: Human Research Program Crew Health. Appendix A&B
Start Date:	07/23/2020	**End Date:**	09/30/2021
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	4
No. of Bachelor's Candidates:	6	**Monitoring Center:**	NASA JSC
Contact Monitor:	Lemery, Jay	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: original end date was 7/22/2021; extended to 9/30/2021 with new PI (Ed., 4/26/21)		
Key Personnel Changes/Previous PI:	NOTE APRIL 2021: SEE ALSO BUCKLAND, DANIEL, MD, PhD. Original Principal Investigator (PI) was Daniel Buckland, MD, PhD; Leila Bridgeman, PhD, worked on the project from the beginning and now is PI due to Dr. Buckland's appointment to Intergovernmental Personnel Agreement, per Exploration Medical Capabilities element of Human Research Program.		
COI Name (Institution):			
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Task Description:	An important step in medical care and research of astronauts involves testing blood or giving intravenous medications. Placing a catheter into a blood vessel takes some degree of training and takes up crewmember's time. The goal of this project is to develop an automated technology that can identify blood vessels (such as a vein or artery), place a needle into the blood vessel, and then place a catheter into the blood vessel so that blood samples can be removed, and medications administered. The technology will operate under human supervision, but not require any active human control, to reduce the amount of training needed to operate it. The project will use a robotic arm with an attached ultrasound probe to automatically identify blood vessels using computer vision techniques, and then use that same robotic arm to insert the needle and catheter. Initial work will be done in simulations of human arms and then progress to human testing when determined safe. The significance of this work could lead to improved medical care of astronauts, as it will no longer require as many crewmembers or resources to start medical diagnosis and treatment. It could also allow for more complete automation or remote operation of medical care for astronauts. This technology could eventually help start medical care for patients on the ground who live in remote locations or do not have enough medical staff available.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Extracting blood is key to myriad medical tests while IV insertion is crucial to administering numerous medicines. Consequently, automated vascular access is a necessary first step in creating virtually any multi-use, autonomous robotic platform. Such technologies could eventually help start medical care for patients on the ground who live in remote locations or do not have enough medical staff available. Even on its own, automated vascular access can unburden medical staff in busy locations. Furthermore, ultrasound-based identification can mitigate the challenges in locations where staff lack practice identifying vasculature of people with varying skin color, or when they must identify vasculature deeper below the surface. Finally, the automated ultrasound tracking implemented for vascular wall identification is a first step towards more robust automated ultrasound scanning that can be applied to a variety of body parts, enabling numerous procedures based on diagnostic scans.
	Aim 1: Vein Target Development This aim proposed to develop an autonomous ultrasound vascular imaging system that uses robotic manipulation of an ultrasound probe to generate 3D vascular wall anatomy model. All milestones of this aim have been completed: a. Develop and test robotic system to maintain constant normal force while tracking trajectories: The system maintained normal contact in tests with a phantom tissue surface by using an RGB-D camera to generate a surface which is used to define trajectories. The force of the ultrasound attached to the robotic arm was adapted based on end-effector force readings. b. Develop and test system to generate vascular wall anatomy model: The vascular wall anatomy model required a maximum root-mean-square error of the centroid along the longitudinal axis of less than 1 mm. In tests with phantom tissue, the vascular anatomy wall model achieved +/-0.3 mm precision root mean square error ,which is comprised of robot arm error (0.03 mm), the error of ultrasound (~0.25 mm), and errors in our vision algorithm. To begin, all point cloud data into the robot base frame coordinates. The location of the vasculature is also defined in this frame, allowing us to visualize the phantom arm surface from RGB-D camera and vessel centroid locations in 3D space. Once the point cloud has been created to identify the arm surface to be scanned, the system takes a depth image of the target surface used to generate the trajectory position and normal. Second, the trajectory is either explicitly chosen, as is done for precision tests, or automatically placed on the surface, as is done in our demonstration. In either case, a splined trajectory of positions and normal directions is generated from the raw depth data. As the robot moves across the surface, the force at the end effector is used to adjust the next position along the trajectory in the normal direction based on a PID (proportional–integral–derivative) controller. Our system had a target force of 3.5N and maximum safety threshold force of 7N. While the robot follows the predetermined trajectory, ultrasound images are acquired and analyzed in real time for initial detection using U-net and then tracked using an active contour-Kalman filter to collect the centroids of the vessel. The success rate was around 97% for both with the errors due to insufficient contact of the probe with the surface. Processing time was 12 fps for identification and 32 fps for tracking, basically in real time for our procedure. Finally, this data is transformed to generate the vessel locations in the robot base frame coordinates. In tests, the variance in the vessel centroid location was approximately 0.3 mm, well below the target vasculature's diameter of 5 mm. This demonstrates the effectiveness of the proposed system for vascular wall model development and tracking. Aim 2: Needle Insertion/Catheter Deployment This aim proposed to develop a robotic mechanism to autonomously guide a needle tip into a target vessel and deploy a peripheral catheter with ultrasound guidance. To achieve our target precision and cannulate a vein, there must be: accurate rotational motion to target the needle effectively; accurate linear motion to place the needle in the target centroid; and secure, minimal fiction joints to connect these aspects to themselves and the current ultrasound end effector. Mechanisms to achieve the first two of these maneuvers have been developed and constructed. The placement and rotation mechanisms achieved the required accuracy in separate tests, but the required accuracy was not achieved in integrated testing, due to slippage of the key gear connecting the motor to the rotating parallelogram. Slippage between gears was significantly reduced by the addition of a lightweight aluminum plate and reducing force by the addition of torsional springs, but the screw locking the large gear to the shaft was stripped because of initial large loads placed on it, requiring replacement for the gear in limited supply (at least 3 months to acquire prior to additional shipping issues of the past year). The catheter deployment mechanism was designed and tested with the phantom and shown to provide sufficient force to keep the catheter in place when the needle was retracted. Progress of the milestones is described below. a. Needle placement mechanism development and testing: The robotic manipulator can autonomously guide a needle tip to within 0.5 mm of a target centroid in a simulated vein of approximately 4 mm diameter. Testing of the needle and catheter system confirms that the motors controlling the needle distance into a target vein can achieve 0.5 mm accuracy. b. Angle system development and testing: The needle must be rotated from the large angle (approximately 40 degrees)

Task Progress:

that is ideal for vein puncture to a smaller angle (as close to 0 degrees as possible) that is better to thread the catheter into the vein. The current prototype design allows angles 10-35 degrees, which is sufficient for insertion and threading the catheter into the vasculature. Final design allows this rotation, but as noted, the gear slippage does not allow final testing.

c. Potential integration of robotic arm, ultrasound, and needle/catheter: Initial tests of the integrated needle placement and rotation systems have been completed, but the required accuracy was not achieved. Consequently, the needle insertion and rotation devices were remodeled to reduce gear slippage and decrease load on the main rotation motor using 8 torsional springs. Final tests of the linear motors with needle and catheter have confirmed 0.3 mm accuracy with the embedded encoders. Encoder opposite the motor on the large rational mechanism can detect accurately rotations of 0.2 degrees. As noted, final tests of the integrated system could not be completed due to large gear slippage.

In the prototype, the needle extends for insertion at close to 30 degrees and then the rotational aspect changes the angle to 10 degrees while the needle tip remains in the vein. After this change in angle, the catheter motor extends the catheter as the needle motor retracts the needle simultaneously. Finally, the catheter will be left in the arm for final securement by the human user or controller.

Linear motion of the needle insertion and catheter release, propelled by linear actuator motors by Actuonix, have a tested accuracy of 0.3 mm, similar to their specifications from the manufacturer. The rotational aspects of the system have been designed and tested with the linear mechanism, but there is unresolvable slack in the integrated system due to the main gear slipping on the shaft that would need to be resolved. Sensing of both the motor itself and encoder could be used to ensure no remaining slack is left in the system and the whole parallelogram move smoothly and accurately. In the refined prototype, constraints on the shafts were used to reduce slippage, and a metal plate was being used to maintain structural integrity, while respecting the weight limit of the end effector of the UR5e robot arm being used.

The final development and testing has started with vascular targets defined within the same phantom from Aim 1. Ultrasound along the longitudinal axis of the vasculature should be used to confirm the needle insertion. Needle placement in the vessel can be confirmed with the blood flash visible in the needle chamber, either automatically with a color sensor or by an inexpert human supervisor. Efficacy of both options are explored with deliverables from Aim 3, where aspects of human robot interaction are being explored, in human testing starting this Spring 2022. [Ed. note: human testing is funded internally from Duke University.]

Aim 3: Human Robot Interaction

This aim proposed to develop a user interface and safety controls enabling personnel with limited training to sagely supervise autonomous catheterization. To achieve this, we are developing an arm restraint to ensure the vessel centroid, which is the target for the needle placement system, remains still. We are also developing an interface for human supervisors to assess whether the vessel has been detected and the needle has been deployed correctly. Clinician feedback was collected to ensure the designs are effective and fit within a medical workflow. This encompassed the following milestones:

a. Develop arm restraint for human subjects: A safety mechanism is needed to secure the patient's arm remains in place and so that the vessel centroid remains still within tolerance of the imaging and guidance system. The first prototype was made for arm securement with air ultrasound sensors to separately test for x, y, and z movement. A second version was developed to secure the patient arm with full access to the wrist through elbow for the procedure while still detecting movement within the restraint. Experiments with human subjects are being planned for this spring to validate the effectiveness of the new design.

b. Develop interfaces for human supervision and acquire clinician input: Test visualizations have been developed from data collected during the procedure in Aim 1 for use in the human robotic interface. Additionally, clinicians have been observed and timed to best understand the steps that would most benefit from human intervention or oversight. The human interface for the needle insertion and catheter placement and securement has gone through multiple iterations, to be finalized before Institutional Review Board (IRB) approved human testing this Spring 2022. [Ed. note: human testing is funded internally from Duke University.]

The proposed arm restraint must be both safe and comfortable for the user. It must accurately detect and alert the human supervisor of movement that would affect the efficacy of the system and require a restart. This is critical as the RGB-D image taken at the beginning of the intervention is used throughout the procedure and excessive movement would make the initial surface collected incorrect, affecting the normal directions and positions used in the trajectory. The final prototype maximizes surface area for the RGB-D camera to collect data and ultrasound to search safely and effectively for target veins.

The final design has been chosen with two key improvements: securing the palm and keeping the elbow in a straight, rather than bent position, will reduce patient motion and allow the surface of the arm to be completely accessible. Restraining the arm and forearm in this form will accommodate patients of different arm length with easily self adjustable telescoping aspects for the hand placement and forearm support.

The arm is held straight and not bent to maximize the surface area, with the restraints limited to the hand and bicep--done in part with the incorporated pressure cuff which will both take pressure and act as tourniquet during the procedure. The hand tightening device can be done by the user themselves with a knob much closer the body.

Additionally, the arm restraint system had initial testing for efficacy with incorporated sensors that ensure the patient has not moved during the procedure. Multiple tests have been performed to verify efficacy of the sensors and the restraint systems, with noise of the sensors of about 3-4 mm.

Another aspect of the arm restraint is a blood pressure cuff. This cuff was developed with automatically adjustable pressure to serve as a tourniquet in the IV placement procedure. Future work will use it to provide typical blood pressure data in conjunction with the ultrasound from the robotic arm to collect data for autonomously determining arteries versus veins. Initial autonomous vasculature identification models have been populated with initial data, which will be supplemented by the human study this spring that will also collect force data in this aspect of the procedure. Data collection during the spring experiment will include placing a select amount of pressure based on BP (blood pressure) measurements on the target vasculature to occlude veins without occluding arteries. This is one aspect of the procedure we are exploring the benefits of untrained human versus robotic decision making.

Next, we focus on the human robotic interface for the IV procedure. Through our work in Aim 1, we have created visualizations for the real time capture of vessel locations. We showed different versions of these visualizations and raw ultrasound images to clinicians to collaboratively better understand any error that could happen the robotic system and how untrained users could be used to catch them. This led to identifying a key question for untrained users: is the target vessel for the procedure linear enough for insertion? This may not be the case for branching targets. We have an example of this branching in our current phantom to use for these tests. Critically, we implemented the stop/go timesteps decided from clinician observation as described in the proposal. We also developed an interface to teach users to secure the catheter once it has been placed autonomously. This is important as cleaning the area is critical to avoid infection and securing the catheter immediately post robotic procedure will ensure consistent vascular access.

Finally, we acquired feedback from clinicians on interface design and usability. Their feedback has been critical for any human testing as they are most aware of the risks and uncertainties of the procedure. Key questions included: Are these critical steps the only places it makes sense to have stop/go pauses in the procedure? Are these securements sufficient and safe for patients to use on a regular basis? Do you trust patients to help make decisions in different aspects of this procedure with the information given, such as: to identify branching, vein v. artery with pressure difference shown, and needle and catheter placement? Would you trust this system and interface to safely perform IV insertions? If not, why and what aspects of the interface could be improved? Target clinicians to receive feedback from included emergency medicine physicians and ultrasound IV trained nurses.

Key feedback received included the importance of user testing for the securement system and objectively quantifying the ability of participants to effectively secure themselves. This input has shaped our coming human testing plan. Ultrasound distance sensors are used during the procedure to ensure users are still and final data from these sensors after the human experiment this spring will validate how effective the securement was. Additionally, researchers will note any additional input needed to ensure participants are secure during the IRB approved human experiment.

Clinicians agreed that the stopping places for the procedure were well placed, but one clinician noted that they did not believe non-clinicians could effectively evaluate if the target vasculature were accurately chosen. This will be evaluated in the human experiments with a post procedure survey that will assess what types of errors non-clinicians could evaluate. This information would be used along with accuracy data for the autonomous identification and tracking to evaluate the safety and efficacy of the complete system. Additionally, clinicians noted the presence of flash (red blood in the tube behind the needle) would be the best final indicator of effective cannulation of the vein versus the ultrasound visual of the autonomous system with the assistance of the human to assess if the target vein has been cannulated. In future iterations of the device, the addition of a flash sensor could be added to automate this final check. Additionally, in practice, if these aspects could be seen by the future user, they could do the final check visually. [Note: needle placement and cannulation are not part of the IRB approved human experiment this spring, funded internally by Duke University.]

Finally, clinicians noted they would trust the system to place IVs if all safety and efficacy objectives for the system are met, but one clinician noted the look of the robotic arm with end effector, especially with the needle system, would take significant adjustment for users, clinical or otherwise -- and in the future might benefit from a different form factor. Participants in the human study this spring will be asked for their trust in the system to perform the task before and after the experiment with just ultrasound searching on the forearm and these responses will help inform the future iterations of the design and procedure.

Bibliography Type:	Description: (Last Updated: 02/02/2022)
Abstracts for Journals and Proceedings	Oca SR, Ma G, Bridgeman L, Buckland DM. "A Robotic System for Ultrasound Guided Peripheral Vascular Localization for Human Spaceflight." 2021 NASA Human Research Program Investigators' Workshop, Virtual, February 1-4, 2021. Abstracts. 2021 NASA Human Research Program Investigators' Workshop, Virtual, February 1-4, 2021. , Feb-2021
Papers from Meeting Proceedings	Ma G, Oca SR, Zhu Y, Codd P, Buckland DM. "A Novel Robotic System for Ultrasound-Guided Peripheral Vascular Localization." 2021 IEEE International Conference on Robotics and Automation (ICRA), X'ian, China, May 30-June 5, 2021. 2021 IEEE International Conference on Robotics and Automation (ICRA), X'ian, China, May 30-June 5, 2021. https://doi.org/10.1109/ICRA48506.2021.9561924 , Jun-2021